

Physical ER Plan — AITasker MVP (PostgreSQL · 28 Tables)

Global Conventions

- **All PKs:** `UUID NOT NULL DEFAULT gen_random_uuid()`
- **All timestamps:** `TIMESTAMPTZ` (UTC)
- **All money:** `BIGINT` (VND integer — no decimals)
- **All JSON:** `JSONB`
- **Enums:** `TEXT` with `CHECK` constraints
- **Extension:** `CREATE EXTENSION IF NOT EXISTS "pgcrypto";`

Section 1 — Users & Role Profiles

Table: `users`

Column	Type	Constraints
<code>id</code>	UUID	PK NOT NULL DEFAULT <code>gen_random_uuid()</code>
<code>email</code>	TEXT	NOT NULL UNIQUE
<code>password_hash</code>	TEXT	NOT NULL
<code>full_name</code>	TEXT	NOT NULL
<code>phone</code>	TEXT	NULL
<code>roles</code>	JSONB	NOT NULL DEFAULT '[]'
<code>active_role</code>	TEXT	NOT NULL CHECK (<code>active_role</code> IN ('CLIENT','EXPERT','ADMIN'))
<code>client_subtype</code>	TEXT	NULL CHECK (<code>client_subtype</code> IN ('CEO','TECH_TEAM'))
<code>subscription_client_tier</code>	TEXT	NOT NULL DEFAULT 'free' CHECK (<code>subscription_client_tier</code> IN ('free','pro'))
<code>subscription_expert_tier</code>	TEXT	NOT NULL DEFAULT 'free' CHECK (<code>subscription_expert_tier</code> IN ('free','pro'))
<code>sub_client_expires_at</code>	TIMESTAMPTZ	NULL
<code>sub_expert_expires_at</code>	TIMESTAMPTZ	NULL
<code>sepay_bank_account_xid</code>	TEXT	NULL
<code>bank_account_holder_name</code>	TEXT	NULL
<code>bank_linked_at</code>	TIMESTAMPTZ	NULL
<code>self_technical</code>	BOOLEAN	NOT NULL DEFAULT FALSE
<code>is_active</code>	BOOLEAN	NOT NULL DEFAULT TRUE
<code>created_at</code>	TIMESTAMPTZ	NOT NULL DEFAULT <code>now()</code>

Indexes: `email` (unique), `active_role`, `client_subtype`

Table: `client_profiles`

Column	Type	Constraints
<code>user_id</code>	UUID	PK NOT NULL REFERENCES <code>users(id)</code> ON DELETE CASCADE
<code>company_name</code>	TEXT	NULL
<code>industry</code>	TEXT	NULL
<code>ceo_name</code>	TEXT	NULL

Table: `expert_profiles`

Column	Type	Constraints	Notes
<code>user_id</code>	UUID	PK NOT NULL REFERENCES <code>users(id)</code> ON DELETE CASCADE	
<code>bio</code>	TEXT	NULL	
<code>engagement_model</code>	TEXT	NULL CHECK (<code>engagement_model</code> IN ('MILESTONE','HOURLY','HYBRID'))	

Column	Type	Constraints	Notes
stack_tags_json	JSONB	NOT NULL DEFAULT '[]'	Absorbs cut expert_stack_tags table. Format: ["Python", "Kafka", "Go"]
archetype_history_json	JSONB	NULL	Self-declared history for cold-start matching. Format: [{archetype_code, tier, self_declared: true}]

Table: tech_team_profiles

Column	Type	Constraints	Notes
user_id	UUID	PK NOT NULL REFERENCES users(id) ON DELETE CASCADE	
linked_client_id	UUID	NOT NULL REFERENCES users(id)	CEO who issued the handoff link
linked_project_id	UUID	NOT NULL REFERENCES projects(id)	Scope guard: account locked to one project
role_title	TEXT	NULL	

Index: linked_project_id

Section 2 — Wallet & Finance

Table: wallets

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
user_id	UUID	NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE
available_balance	BIGINT	NOT NULL DEFAULT 0 CHECK (available_balance >= 0)
locked_balance	BIGINT	NOT NULL DEFAULT 0 CHECK (locked_balance >= 0)

Seeded system wallet on migration:

```
INSERT INTO users (id, email, password_hash, full_name, roles, active_role, is_active, created_at)
VALUES ('00000000-0000-0000-0000-000000000001', 'platform@aitasker.internal',
'SEEDED_NO_LOGIN', 'AITasker Platform', ['ADMIN'], 'ADMIN', true, now());
INSERT INTO wallets (user_id) VALUES ('00000000-0000-0000-0000-000000000001');
-- Store returned wallet id as env var PLATFORM_FEE_WALLET_ID
```

Table: wallet_transactions (immutable ledger)

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
wallet_id	UUID	NOT NULL REFERENCES wallets(id)
amount	BIGINT	NOT NULL CHECK (amount > 0)
transaction_type	TEXT	NOT NULL CHECK (transaction_type IN ('TOP_UP','SUBSCRIPTION','ESCROW_LOCK','ESCROW_RELEASE','PLATFORM_FEE','ESCROW_REFUND','ESCROW_SPLIT','WITHI
reference_id	TEXT	NULL
created_at	TIMESTAMP TZ	NOT NULL DEFAULT now()

Idempotency index (prevents SePay retry double-credit):

```
CREATE UNIQUE INDEX wallet_tx_idempotency
ON wallet_transactions(wallet_id, reference_id)
```

```
WHERE reference_id IS NOT NULL;
```

Table: virtual_accounts

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
entity_type	TEXT	NOT NULL CHECK (entity_type IN ('WALLET_TOPUP','MILESTONE','SERVICE','SUBSCRIPTION'))	
entity_id	TEXT	NOT NULL	Polymorphic: user_id / milestone_id / engagement purchase_ref / user_subscriptions.id
va_number	TEXT	NOT NULL UNIQUE	SePay-issued bank account number
fixed_amount	BIGINT	NULL	NULL for WALLET_TOPUP; set for all others
expires_at	TIMESTAMPTZ	NULL	24h for MILESTONE VAs; NULL for WALLET_TOPUP
status	TEXT	NOT NULL DEFAULT 'ACTIVE' CHECK (status IN ('ACTIVE','EXPIRED','USED'))	

Table: withdrawal_requests

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
expert_id	UUID	NOT NULL REFERENCES expert_profiles(user_id)	
type	TEXT	NOT NULL DEFAULT 'EXPERT_MANUAL' CHECK (type IN ('MILESTONE_RELEASE','EXPERT_MANUAL'))	MILESTONE_RELEASE auto-created on APPROVED; EXPERT_MANUAL expert-initiated
amount	BIGINT	NOT NULL CHECK (amount > 0)	
bank_account_xid	TEXT	NOT NULL	
disbursement_id	TEXT	NULL	SePay chi hộ response ID
status	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING','PROCESSING','COMPLETED','FAILED'))	
requested_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	
confirmed_at	TIMESTAMPTZ	NULL	

Table: platform_settings (singleton)

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
platform_wallet_id	UUID	NOT NULL UNIQUE REFERENCES wallets(id)
platform_fee_pct	FLOAT	NOT NULL DEFAULT 0.05 CHECK (platform_fee_pct BETWEEN 0 AND 1)
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Section 3 — Elicitation Engine

Table: elicitation_sessions

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
user_id	UUID	NOT NULL REFERENCES users(id)	CEO who started the session
current_stage	INT	NOT NULL DEFAULT 1 CHECK (current_stage BETWEEN 1 AND 5)	
archetype	TEXT	NULL	Locked after Stage 2
scenario_type	TEXT	NULL CHECK (scenario_type IN ('STANDARD','SCENARIO_A','SCENARIO_B'))	
void_list_json	JSONB	NOT NULL DEFAULT '[]'	[{void_code, injected: boolean}]
state	TEXT	NOT NULL DEFAULT 'IN_PROGRESS' CHECK (state IN ('IN_PROGRESS','COMPLETED','ABANDONED','RETURNED'))	

Column	Type	Constraints	Notes
<code>created_at</code>	TIMESTAMPZ	NOT NULL DEFAULT now()	
<code>updated_at</code>	TIMESTAMPZ	NOT NULL DEFAULT now()	

Section 4 — Projects (JSONB Hub)

Table: `projects`

Column	Type	Constraints	Notes
<code>id</code>	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
<code>client_id</code>	UUID	NOT NULL REFERENCES users(id)	CEO user_id
<code>elicitation_session_id</code>	UUID	NULL REFERENCES elicitation_sessions(id)	NULL for non-elicitation paths
<code>state</code>	TEXT	NOT NULL DEFAULT 'DRAFT' CHECK (state IN ('DRAFT','PUBLISHED','RETURNED_TO_CLIENT','SUSPENDED'))	
<code>archetype</code>	TEXT	NULL	
<code>tier</code>	TEXT	NULL CHECK (tier IN ('TIER_1','TIER_2','TIER_3'))	
<code>self_technical</code>	BOOLEAN	NOT NULL DEFAULT FALSE	
<code>required_seams_json</code>	JSONB	NOT NULL DEFAULT '[]'	Replaces <code>capability_footprints.required_seams_json</code>
<code>required_domains_json</code>	JSONB	NOT NULL DEFAULT '[]'	Replaces <code>capability_footprints.required_domains_json</code>
<code>milestone_framework_json</code>	JSONB	NOT NULL DEFAULT '[]'	Replaces <code>capability_footprints.milestone_framework_json</code>
<code>artifact_a_json</code>	JSONB	NULL	Replaces <code>artifact_a</code> table. Visible to matched e: pre-bid
<code>artifact_b_json</code>	JSONB	NULL	Replaces <code>artifact_b</code> table. FastAPI guard: only r when <code>engagement.state >= CONNECTED AND N both accepted AND requester != CEO</code>
<code>created_at</code>	TIMESTAMPZ	NOT NULL DEFAULT now()	

Index: `client_id, state`

Section 5 — Services

Table: `services`

Column	Type	Constraints
<code>id</code>	UUID	PK NOT NULL DEFAULT gen_random_uuid()
<code>expert_id</code>	UUID	NOT NULL REFERENCES expert_profiles(user_id) ON DELETE CASCADE
<code>title</code>	TEXT	NOT NULL
<code>description</code>	TEXT	NOT NULL
<code>domains_json</code>	JSONB	NOT NULL DEFAULT '[]'
<code>seams_json</code>	JSONB	NOT NULL DEFAULT '[]'
<code>price_vnd</code>	BIGINT	NOT NULL CHECK (price_vnd > 0)
<code>state</code>	TEXT	NOT NULL DEFAULT 'DRAFT' CHECK (state IN ('DRAFT','PUBLISHED','SUSPENDED'))
<code>service_type</code>	TEXT	NOT NULL CHECK (service_type IN ('AI_SERVICE','TECH_DISCOVERY'))

Section 6 — Expert Capability (2-Tier)

Table: `expert_domain_depths`

Column	Type	Constraints
<code>id</code>	UUID	PK NOT NULL DEFAULT gen_random_uuid()
<code>expert_id</code>	UUID	NOT NULL REFERENCES expert_profiles(user_id) ON DELETE CASCADE

Column	Type	Constraints
domain_code	TEXT	NOT NULL
depth_level	TEXT	NOT NULL CHECK (depth_level IN ('SURFACE','OPERATIONAL','DEEP'))
verification_tier	TEXT	NOT NULL DEFAULT 'CLAIMED' CHECK (verification_tier IN ('CLAIMED','EVIDENCE_BACKED'))

Unique constraint: (expert_id, domain_code)

Table: expert_seam_claims

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
expert_id	UUID	NOT NULL REFERENCES expert_profiles(user_id) ON DELETE CASCADE	
seam_code	TEXT	NOT NULL	
verification_tier	TEXT	NOT NULL DEFAULT 'CLAIMED' CHECK (verification_tier IN ('CLAIMED','EVIDENCE_BACKED'))	Tier 2 only for MVP
submission_count	INT	NOT NULL DEFAULT 0	5-attempt throttle
locked_until	TIMESTAMPTZ	NULL	

Unique constraint: (expert_id, seam_code)

Table: portfolio_submissions

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
expert_id	UUID	NOT NULL REFERENCES expert_profiles(user_id) ON DELETE CASCADE	
seam_claim_id	UUID	NOT NULL REFERENCES expert_seam_claims(id)	Which claim is being upgraded to Tier 2
project_description	TEXT	NOT NULL	
decision_points	TEXT	NOT NULL	Structured rubric response
status	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING','APPROVED','REJECTED'))	
llm_confidence	FLOAT	NULL	Threshold ≥ 0.85 for auto-upgrade
submitted_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	
evaluated_at	TIMESTAMPTZ	NULL	

Section 7 — Engagements

Table: engagements

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
project_id	UUID	NULL REFERENCES projects(id)	NULL for Path B
expert_id	UUID	NOT NULL REFERENCES expert_profiles(user_id)	
service_id	UUID	NULL REFERENCES services(id)	NULL for Path A
type	TEXT	NOT NULL CHECK (type IN ('PROJECT_BASED','SERVICE_PURCHASE','TECH_DISCOVERY'))	Immutable after creation
state	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (state IN ('PENDING','CONNECTED','ACTIVE','CLOSED','DISPUTED'))	
connected_at	TIMESTAMPTZ	NULL	
client_nda_accepted_at	TIMESTAMPTZ	NULL	Required before CONNECTED
expert_nda_accepted_at	TIMESTAMPTZ	NULL	Required before CONNECTED

Table-level CHECK (type-FK consistency):

```

CONSTRAINT engagement_type_fk CHECK (
  (type = 'PROJECT_BASED' AND project_id IS NOT NULL AND service_id IS NULL) OR
  (type IN ('SERVICE_PURCHASE', 'TECH_DISCOVERY') AND project_id IS NULL AND service_id IS NOT NULL)
)

```

Section 8 — Capability Bids (Simplified)

Table: `capability_bids`

Column	Type	Constraints
<code>id</code>	UUID	PK NOT NULL DEFAULT <code>gen_random_uuid()</code>
<code>engagement_id</code>	UUID	NOT NULL UNIQUE REFERENCES <code>engagements(id)</code>
<code>footprint_alignment_json</code>	JSONB	NULL
<code>approach_summary</code>	TEXT	NULL
<code>conditional_pricing_json</code>	JSONB	NULL
<code>state</code>	TEXT	NOT NULL DEFAULT 'DRAFT' CHECK (state IN ('DRAFT', 'SUBMITTED', 'TECH_REVIEW', 'REVISION_REQUESTED', 'TECH_APPROVED', 'CEO_REVIEW', 'SELECTED', 'DECLINED'))
<code>tech_status</code>	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (tech_status IN ('PENDING', 'APPROVED', 'REVISION_REQUESTED'))
<code>ceo_status</code>	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (ceo_status IN ('PENDING', 'APPROVED', 'DECLINED'))
<code>tech_feedback</code>	TEXT	NULL
<code>negotiated_price_vnd</code>	BIGINT	NULL
<code>version_number</code>	INT	NOT NULL DEFAULT 1

Section 9 — Milestones (2-Layer)

Table: `milestones`

Column	Type	Constraints
<code>id</code>	UUID	PK NOT NULL DEFAULT <code>gen_random_uuid()</code>
<code>engagement_id</code>	UUID	NOT NULL REFERENCES <code>engagements(id)</code> ON DELETE CASCADE
<code>milestone_number</code>	INT	NOT NULL
<code>deliverable_statement</code>	TEXT	NOT NULL
<code>sign_off_authority</code>	TEXT	NOT NULL CHECK (sign_off_authority IN ('TECH_TEAM', 'CEO', 'JOINT'))
<code>payment_amount_vnd</code>	BIGINT	NOT NULL CHECK (payment_amount_vnd > 0)
<code>state</code>	TEXT	NOT NULL DEFAULT 'DEFINED' CHECK (state IN ('DEFINED', 'AWAITING_PAYMENT', 'FUNDED', 'IN_PROGRESS', 'SUBMITTED', 'IN_REVISION', 'APPROVED', 'RELEASED', 'DISP'))
<code>va_number</code>	TEXT	NULL
<code>va_expires_at</code>	TIMESTAMPTZ	NULL
<code>funded_at</code>	TIMESTAMPTZ	NULL
<code>submitted_at</code>	TIMESTAMPTZ	NULL
<code>approved_at</code>	TIMESTAMPTZ	NULL
<code>released_at</code>	TIMESTAMPTZ	NULL

Unique constraint: (engagement_id, milestone_number)

Table: acceptance_criteria (Layer 1)

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
milestone_id	UUID	NOT NULL REFERENCES milestones(id) ON DELETE CASCADE	
criterion_text	TEXT	NOT NULL	LLM quality-gates on save
is_required	BOOLEAN	NOT NULL DEFAULT TRUE	
verified_by_role	TEXT	NOT NULL CHECK (verified_by_role IN ('TECH_TEAM','CEO','JOINT'))	
verified_at	TIMESTAMPZ	NULL	
revision_note	TEXT	NULL	Absorbs cut revision_requests table. Written by reviewer when rejecting criterion

Table: milestone_dod_items (Layer 2)

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
milestone_id	UUID	NOT NULL REFERENCES milestones(id) ON DELETE CASCADE	
item_description	TEXT	NOT NULL	
is_required	BOOLEAN	NOT NULL DEFAULT TRUE	
status	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING','COMPLETED','NOT_APPLICABLE'))	
completed_at	TIMESTAMPZ	NULL	
completion_note	TEXT	NULL	
not_applicable_note	TEXT	NULL	
maps_to_criterion_id	UUID	NULL REFERENCES acceptance_criteria(id)	Optional upward link

Table-level CHECK:

```
CONSTRAINT dod_required_cannot_be_na
CHECK (NOT (is_required = TRUE AND status = 'NOT_APPLICABLE'))
```

Table: milestone_submissions

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
milestone_id	UUID	NOT NULL REFERENCES milestones(id) ON DELETE CASCADE
expert_id	UUID	NOT NULL REFERENCES expert_profiles(user_id)
description	TEXT	NOT NULL
files_json	JSONB	NOT NULL DEFAULT '[]'
submitted_at	TIMESTAMPZ	NOT NULL DEFAULT now()

Table: paygated_documents (KEPT — core F5 mechanism)

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
milestone_id	UUID	NOT NULL REFERENCES milestones(id) ON DELETE CASCADE	Release trigger: when milestone → FUNDED via IPN
document_url	TEXT	NOT NULL	

Column	Type	Constraints	Notes
release_state	TEXT	NOT NULL DEFAULT 'STAGED' CHECK (release_state IN ('STAGED','RELEASED'))	Flips automatically when IPN fires
staged_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	
released_at	TIMESTAMPTZ	NULL	Set by IPN handler. TECH_TEAM-only read; CEO excluded

Section 10 — Escrow (Dual-Parent)

Table: escrow_accounts

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
milestone_id	UUID	NULL REFERENCES milestones(id)	Path A (PROJECT_BASED)
engagement_id	UUID	NULL REFERENCES engagements(id)	Path B (SERVICE_PURCHASE / TECH_DISCOVERY)
amount	BIGINT	NOT NULL CHECK (amount > 0)	
client_wallet_id	UUID	NOT NULL REFERENCES wallets(id)	Source wallet
expert_wallet_id	UUID	NOT NULL REFERENCES wallets(id)	Destination wallet
status	TEXT	NOT NULL DEFAULT 'HELD' CHECK (status IN ('HELD','RELEASED','FROZEN','REFUNDED','SPLIT'))	
held_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	
released_at	TIMESTAMPTZ	NULL	

Table-level CHECK (exactly one parent):

```
CONSTRAINT escrow_has_one_parent CHECK (
  (milestone_id IS NOT NULL AND engagement_id IS NULL) OR
  (milestone_id IS NULL AND engagement_id IS NOT NULL)
)
```

Two partial unique indexes:

```
CREATE UNIQUE INDEX escrow_milestone_unique
ON escrow_accounts(milestone_id) WHERE milestone_id IS NOT NULL;
CREATE UNIQUE INDEX escrow_engagement_unique
ON escrow_accounts(engagement_id) WHERE engagement_id IS NOT NULL;
```

Section 11 — Disputes (2-Layer)

Table: disputes

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
engagement_id	UUID	NOT NULL REFERENCES engagements(id)	
milestone_id	UUID	NOT NULL REFERENCES milestones(id)	
criterion_id	UUID	NOT NULL REFERENCES acceptance_criteria(id)	Always filed against a specific criterion
escrow_account_id	UUID	NOT NULL REFERENCES escrow_accounts(id)	Direct FK — O(1) freeze, type-agnostic
filed_by	UUID	NOT NULL REFERENCES users(id)	
state	TEXT	NOT NULL DEFAULT 'PENDING' CHECK (state IN ('PENDING','LAYER_1_EVAL','AUTO_RESOLVED','MANUAL_REVIEW','RESOLVED'))	
llm_confidence	FLOAT	NULL CHECK (llm_confidence BETWEEN 0 AND 1)	Layer 1 evaluation
filed_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	

Column	Type	Constraints	Notes
resolved_at	TIMESTAMPTZ	NULL	

Section 12 — Messaging

Table: `messages`

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
engagement_id	UUID	NOT NULL REFERENCES engagements(id) ON DELETE CASCADE
sender_id	UUID	NOT NULL REFERENCES users(id)
content	TEXT	NOT NULL
attachment_url	TEXT	NULL
timestamp	TIMESTAMPTZ	NOT NULL DEFAULT now()

Index: `engagement_id, timestamp`

Table: `message_reads` (*KEPT — F9 unread badge*)

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()
message_id	UUID	NOT NULL REFERENCES messages(id) ON DELETE CASCADE
user_id	UUID	NOT NULL REFERENCES users(id)
read_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Unique constraint: `(message_id, user_id)`

Section 13 — Reviews & Audit

Table: `reviews`

Column	Type	Constraints	Notes
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()	
engagement_id	UUID	NOT NULL REFERENCES engagements(id)	
reviewer_id	UUID	NOT NULL REFERENCES users(id)	
target_id	UUID	NOT NULL REFERENCES users(id)	
rating	INT	NOT NULL CHECK (rating BETWEEN 1 AND 5)	
comment	TEXT	NULL	
structured_signals_json	JSONB	NULL	Absorbs cut <code>outcome_signals.seam_signals_json</code> . Informational only (admin dashboard) since Tier 4 is cut. Format: <code>[{seam_code, signal_type, seam_role}]</code>
reviewer_role	TEXT	NOT NULL CHECK (reviewer_role IN ('CEO','TECH_TEAM','EXPERT'))	

Unique constraint: `(engagement_id, reviewer_id)`

Table: `platform_decisions`

Column	Type	Constraints
id	UUID	PK NOT NULL DEFAULT gen_random_uuid()

Column	Type	Constraints
decision_type	TEXT	NOT NULL CHECK (decision_type IN ('ELICITATION_SYNTHESIS','SPEC_AUTO_RETURN','SEAM_TIER_UPGRADE','PORTFOLIO_EVAL','DISPUTE_L1_EVAL','CRITERION_QL'))
entity_type	TEXT	NOT NULL
entity_id	TEXT	NOT NULL
llm_confidence	FLOAT	NULL CHECK (llm_confidence BETWEEN 0 AND 1)
decision	TEXT	NOT NULL
advisory_note	TEXT	NULL
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

FK Dependency Order (28 tables)

1.	users	
2.	client_profiles	→ users
3.	expert_profiles	→ users
4.	wallets	→ users
5.	wallet_transactions	→ wallets
6.	withdrawal_requests	→ expert_profiles
7.	virtual_accounts	→ (polymorphic – standalone)
8.	elicitation_sessions	→ users
9.	projects	→ users, elicitation_sessions
10.	tech_team_profiles	→ users, projects
11.	platform_settings	→ wallets
12.	services	→ expert_profiles
13.	engagements	→ projects, expert_profiles, services
14.	expert_domain_depths	→ expert_profiles
15.	expert_seam_claims	→ expert_profiles
16.	portfolio_submissions	→ expert_profiles, expert_seam_claims
17.	capability_bids	→ engagements
18.	milestones	→ engagements
19.	acceptance_criteria	→ milestones
20.	milestone_dod_items	→ milestones, acceptance_criteria
21.	milestone_submissions	→ milestones, expert_profiles
22.	paygated_documents	→ milestones
23.	escrow_accounts	→ milestones (nullable), engagements (nullable), wallets
24.	disputes	→ engagements, milestones, acceptance_criteria, escrow_accounts, users
25.	messages	→ engagements, users
26.	message_reads	→ messages, users
27.	reviews	→ engagements, users
28.	platform_decisions	→ (polymorphic – standalone)

Conceptual ER — Unified Draw Plan (MVP · 28 Tables)

Notation:  = Entity (rectangle) ·  = Relationship (diamond) · cardinality on each line

Entities to Draw — 28 Total

users · client_profiles · expert_profiles · tech_team_profiles · wallets · wallet_transactions · virtual_accounts · withdrawal_requests · platform_settings · elicitation_sessions · projects · services · expert_domain_depths · expert_seam_claims · portfolio_submissions · engagements · capability_bids · milestones · acceptance_criteria · milestone_dod_items · milestone_submissions · paygated_documents · escrow_accounts · disputes · messages · message_reads · reviews · platform_decisions

Phase 0 — Elicitation Engine

Draw elicitation_sessions between users and projects.

- users  initiates → elicitation_sessions (1:N)

Phase 1 — Users & Role Subtypes

Draw users as the central anchor.

- users  has → client_profiles (1:1)
- users  has → expert_profiles (1:1)

- `users` ✦ `has` → `tech_team_profiles` (1:1)
- `users` ✦ `invited` → `tech_team_profiles` (1:N) — via `linked_client_id`

Note: No separate `Subscription` entity. `Subscription tier and expiry` are columns on `users`. No separate organization entities — Phase 2.

Phase 2 — Expert Capability (2-Tier System)

All branch from `expert_profiles`.

Note: No `expert_stack_tags` entity — `stack tags` are `stack_tags_json JSONB` on `expert_profiles`. No `scenario_assessments`, `scenario_responses`, or `expert_seam_outcome_signals` — Tiers 3 and 4 are cut.

- `expert_profiles` ✦ `has` → `expert_domain_depths` (1:N)
- `expert_profiles` ✦ `holds` → `expert_seam_claims` (1:N)
- `expert_profiles` ✦ `submits` → `portfolio_submissions` (1:N)
- `portfolio_submissions` ✦ `upgrades` → `expert_seam_claims` (N:1) — via `seam_claim_id`; LLM evaluates and upgrades the targeted claim to Tier 2

Phase 3 — Wallet, Finance & Platform Settings

Draw `wallets` beside `users`. Draw `platform_settings` as a singleton near the wallet cluster.

Note: No `user_subscriptions` entity — `subscription state` lives as columns on `users`. `virtual_accounts` floats near this cluster — no strict FK line since `entity_id` is polymorphic.

- `users` ✦ `owns` → `wallets` (1:1)
- `wallets` ✦ `records` → `wallet_transactions` (1:N)
- `expert_profiles` ✦ `requests` → `withdrawal_requests` (1:N)
- `platform_settings` ✦ `references` → `wallets` (1:1) — annotate: "`singleton` — `seeded at deploy`"

Phase 4 — Projects (JSONB Hub)

Draw `projects` as the primary hub. Annotate the entity box: "`contains artifact_a_json + artifact_b_json + footprint JSONBs` — `replaces 3 cut tables`".

- `users` ✦ `creates` → `projects` (1:N) — CEO only
- `elicitation_sessions` ✦ `produces` → `projects` (1:1) — Stage 5 synthesis output
- `projects` ✦ `scopes` → `tech_team_profiles` (1:N) — via `linked_project_id`

Draw the Artifact B access gate annotation on the `projects` entity: "`artifact_b_json route-gated: state ≥ CONNECTED + NDA accepted + CEO excluded`". No separate Artifact B entity — the gate is a route guard on the same row.

Phase 5 — Services (Path B)

- `expert_profiles` ✦ `creates` → `services` (1:N)

Phase 6 — Engagements & Bids

Draw `engagements` as the secondary hub.

- `projects` ✦ `has` → `engagements` (1:N) — Path A; `project_id` NULL for Path B
- `services` ✦ `generates` → `engagements` (1:N) — Path B only
- `expert_profiles` ✦ `joins` → `engagements` (1:N)

Note: No `spec_clarifications` entity — `pre-bid technical questions` use the `messages` channel.

Draw `capability_bids`:

- `engagements` ✦ `has` → `capability_bids` (1:1) — one bid per engagement; absent for `SERVICE_PURCHASE`

Annotate the `capability_bids` entity box: "`tech_status + ceo_status cols replace 4 surface tables`". No separate `bid_versions`, `bid_revision_requests`, `price_negotiations`, `bid_conflict_overrides` entities.

Phase 7 — Milestones (2-Layer)

Draw `milestones` from `engagements`. Note: Layer 3 (`sprints`, `sprint updates`, `addon phases`) is entirely cut — no `milestone_sprints`, `sprint_status_updates`, `addon_phase_requests`, `sprint_comments`, `dod_item_comments` entities.

Engagement → Milestones:

- `engagements` ✦ `has` → `milestones` (1:N)
- `milestones` ✦ `allocates` → `virtual_accounts` (1:N)

Layer 1 — Acceptance Criteria:

- milestones ⌘ has → acceptance_criteria (1:N) — annotate: "revision_note col replaces revision_requests table"
- acceptance_criteria ⌘ evaluation logged in → platform_decisions (1:1) — LLM quality gate on criterion save

Layer 2 — DoD Checklist:

- milestones ⌘ has → milestone_dod_items (1:N)
- milestone_dod_items ⌘ maps to → acceptance_criteria (N:1) — nullable upward link

Deliverables & Pay-gated Documents:

- milestones ⌘ has → milestone_submissions (1:N)
- expert_profiles ⌘ submits → milestone_submissions (1:N)
- milestones ⌘ triggers release of → paygated_documents (1:N) — annotate: "IPN fires → release_state STAGED→RELEASED; TECH_TEAM-only read"

Escrow (dual-parent paths):

- milestones ⌘ held in → escrow_accounts (1:1) — annotate: "Path A only"
- engagements ⌘ holds escrow for → escrow_accounts (1:1) — annotate: "Path B only (SERVICE_PURCHASE)"
- wallets ⌘ funds → escrow_accounts (1:N) — via client_wallet_id
- wallets ⌘ receives from → escrow_accounts (1:N) — via expert_wallet_id

Disputes (2-Layer):

- engagements ⌘ has → disputes (1:N)
- milestones ⌘ has → disputes (1:N)
- acceptance_criteria ⌘ subject of → disputes (1:N) — disputes always filed against a specific criterion
- escrow_accounts ⌘ frozen by → disputes (1:1) — direct FK, O(1) freeze
- users ⌘ files → disputes (1:N)

Note: No *dispute_resolution_reports* entity — cut.

Phase 8 — Messaging, Reviews & Audit

Messaging:

- engagements ⌘ has → messages (1:N) — one channel absorbs spec clarifications + sprint comments + DoD comments
- users ⌘ sends → messages (1:N)
- messages ⌘ has → message_reads (1:N)
- users ⌘ reads → message_reads (1:N)

Reviews:

- engagements ⌘ has → reviews (1:N) — annotate: "structured_signals_json absorbs cut outcome_signals table"
- users ⌘ writes → reviews (1:N) — via reviewer_id
- users ⌘ reviewed as → reviews (1:N) — via target_id

Platform Decisions:

platform_decisions is a standalone entity with two incoming lines from Phase 7 (acceptance_criteria → evaluation logged in). No further FK lines — entity_id is polymorphic.

Final Count

Phase	Entities	Relationships
0 — Elicitation	1	1
1 — Users & Subtypes	4	4
2 — Expert Capability	3	4
3 — Wallet & Finance	4	4
4 — Projects Hub	1	3
5 — Services	1	1
6 — Engagements & Bids	2	4
7 — Milestones, Escrow, Disputes	10	18
8 — Messaging, Reviews, Audit	4	7
Total	28	46